

Components of Executive Functioning Involved in Prospective Memory Performance



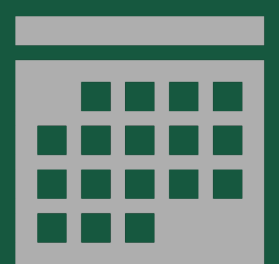
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Introduction

- Prospective memory (PM) is the ability to create and successfully perform future tasks and goals¹.
- Executive functioning (EF) is critical for managing attentional resources required for PM, as EF skills capture planning, monitoring, and cognitive control².
- Previous research has used the Dysexecutive Questionnaire (DEX) to assess EF and divide it into 5 cognitive factors³:
 - Intentionality
 - Interference Management
 - Inhibition
 - Planning
 - Social Regulation
- The preparatory attentional and memory processes (PAM) model suggests that EF abilities must be active before the presentation of a cue to perform an action⁴.
- Given the complex nature of EF, various cognitive factors involved in EF may play different roles in one's ability to successfully engage in PM⁵.
- Inhibition and interference management may play particularly important roles in PM; these cognitive factors allow a person to suppress distractors and competing tasks, facilitating their performance of a future-oriented goal^{6,7}.



The Current Study

- Participants reported EF abilities and completed a PM Task.
- Goal:** To determine which EF components are involved in PM performance.

We hypothesized that impairments in inhibition and interference management would be inversely related to PM accuracy.

Methods



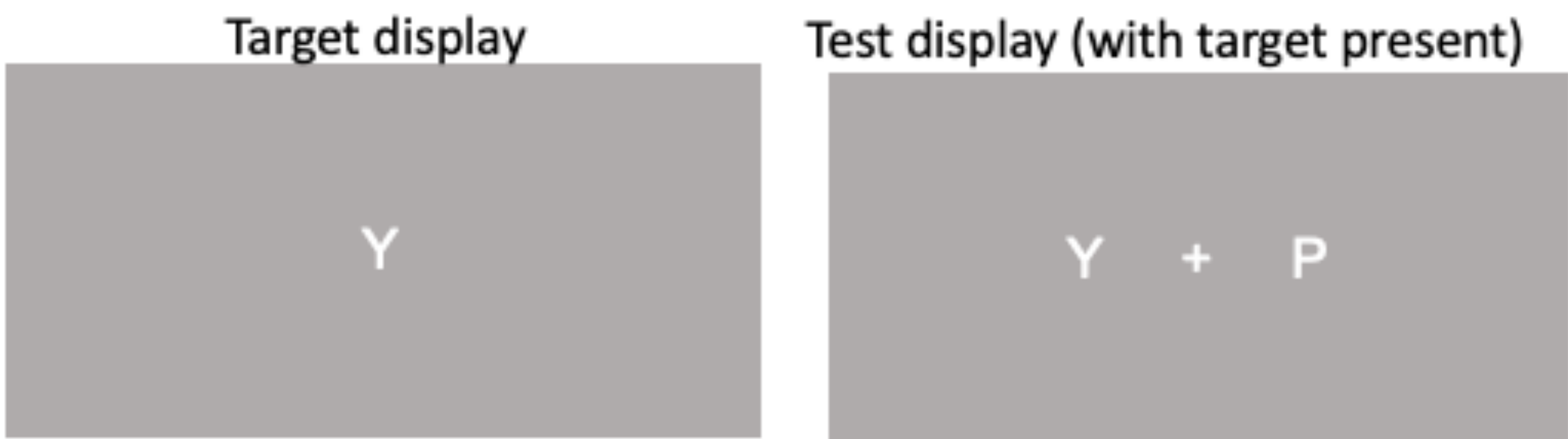
- Participants reported EF abilities via the DEX, which was divided into: Intentionality (F1), Interference Management (F2), Inhibition (F3), Planning (F4), & Social Regulation (F5)².
- Each factor consisted of 3 DEX questions, scored using a 5-point scale according to frequency from “Never” (1) to “Very Often” (5)².
- Example DEX statements:
 - Item 2: “I act without thinking, doing the first thing that comes to mind.”
 - Item 4: “I have difficulty thinking ahead or planning for the future.”

Participants

- 79 undergraduate students (56 women, 17 men, 6 transgender/nonbinary), mean age 20.14 (range 18-35).

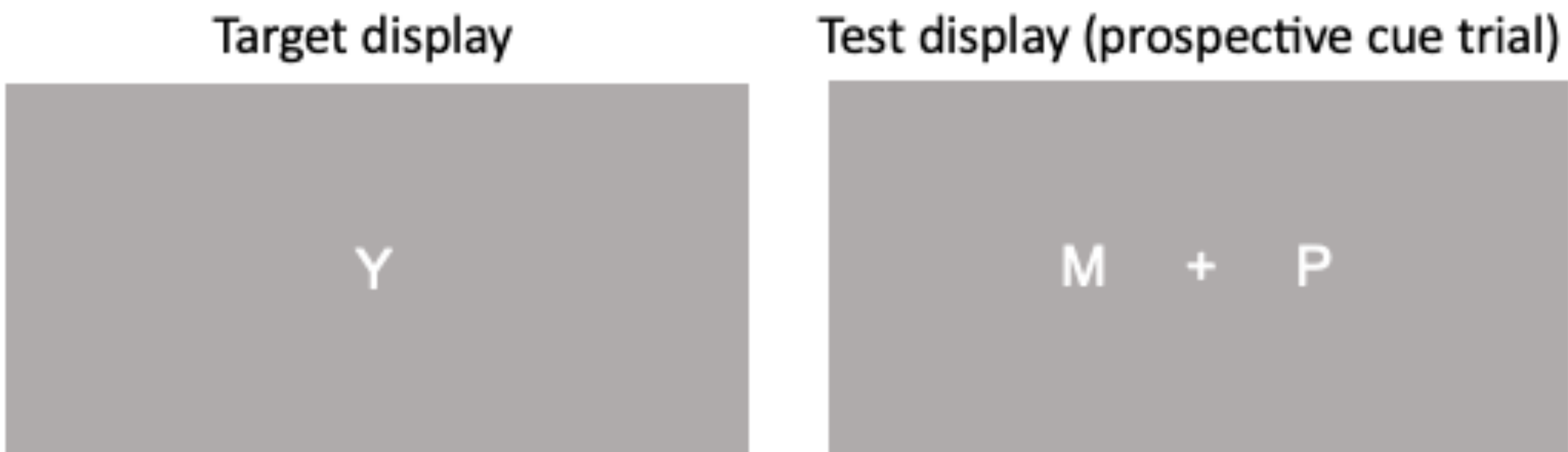
Methods

Task: Ongoing Visual Search Block



Select the V key if the target is present.
Select the B key if the target is absent.

Task: Prospective Memory Block



Select the V key if the target is present.
Select the B key if the target is absent.
Select the N key if D or M is present.

Results

Table 1: Descriptives for DEX Factors & PM Performance

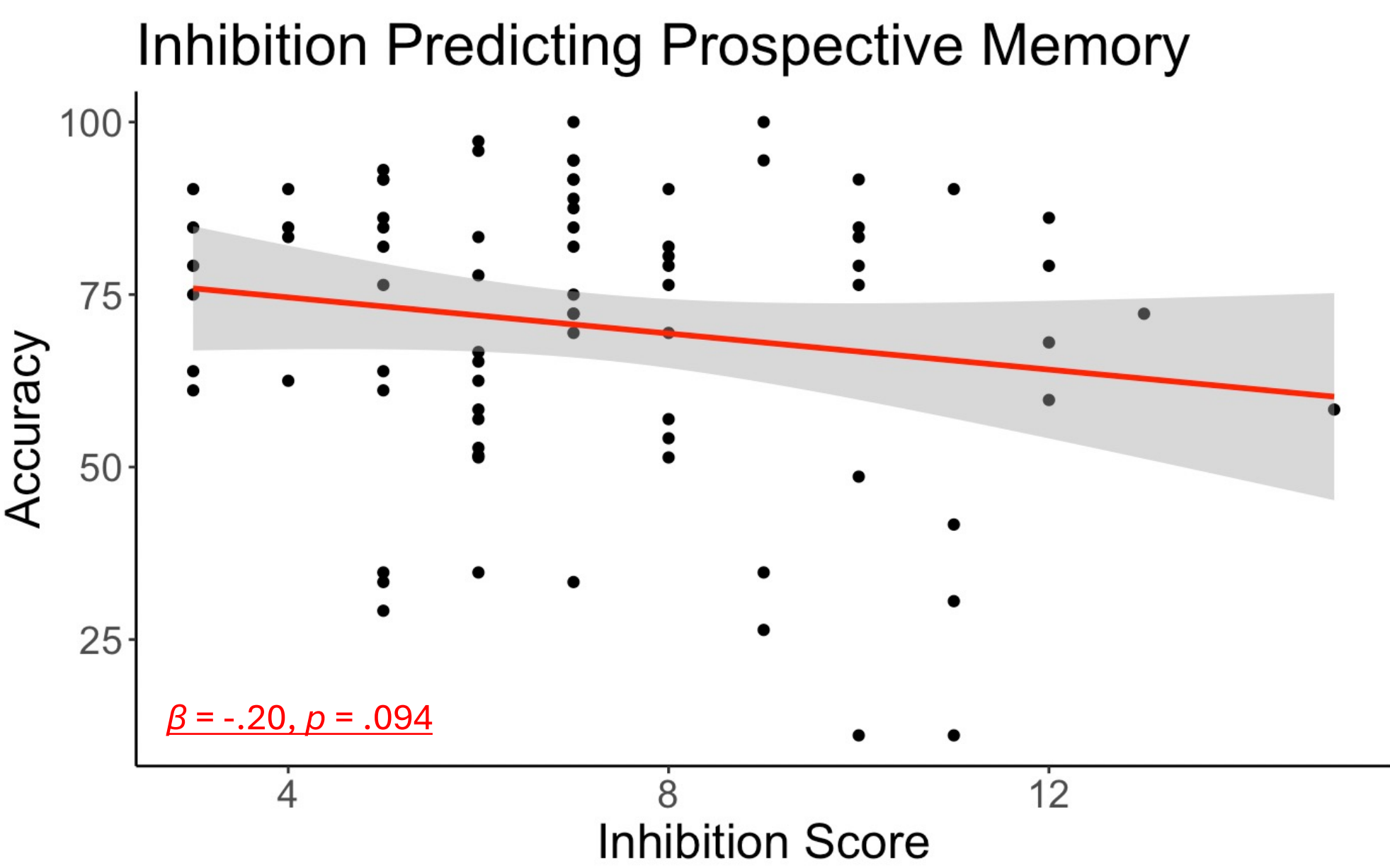
Variable	M	SD
PM Accuracy	70.41%	21.44
F1: Intentionality	6.94	2.37
F2: Interference Management	8.59	2.88
F3: Inhibition	7.19	2.64
F4: Planning	7.87	2.79
F5: Social Regulation	6.49	2.06

Higher scores on each factor indicate a higher degree of cognitive impairment.

- The maximum score for each factor was 15 and the minimum was 3.

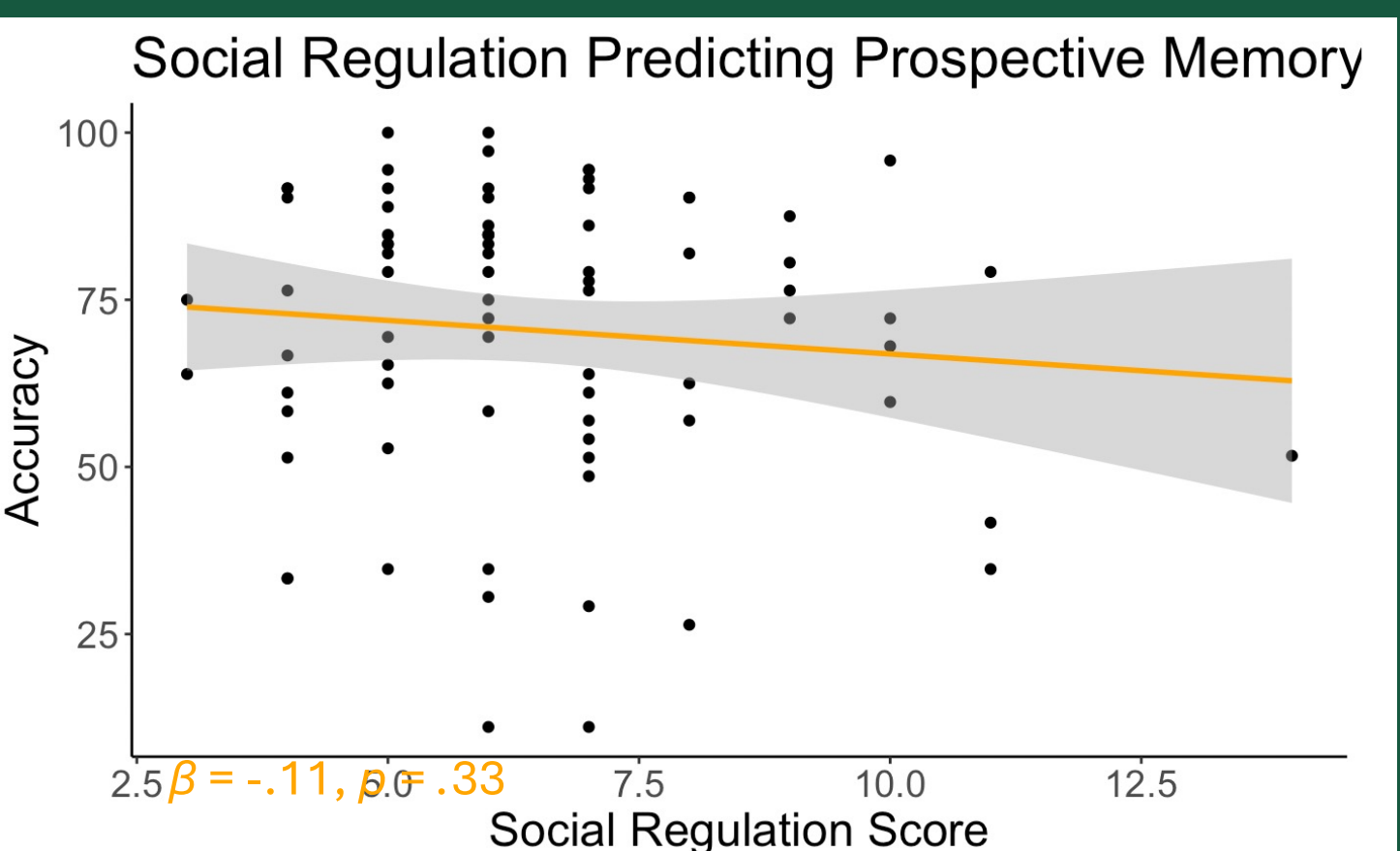
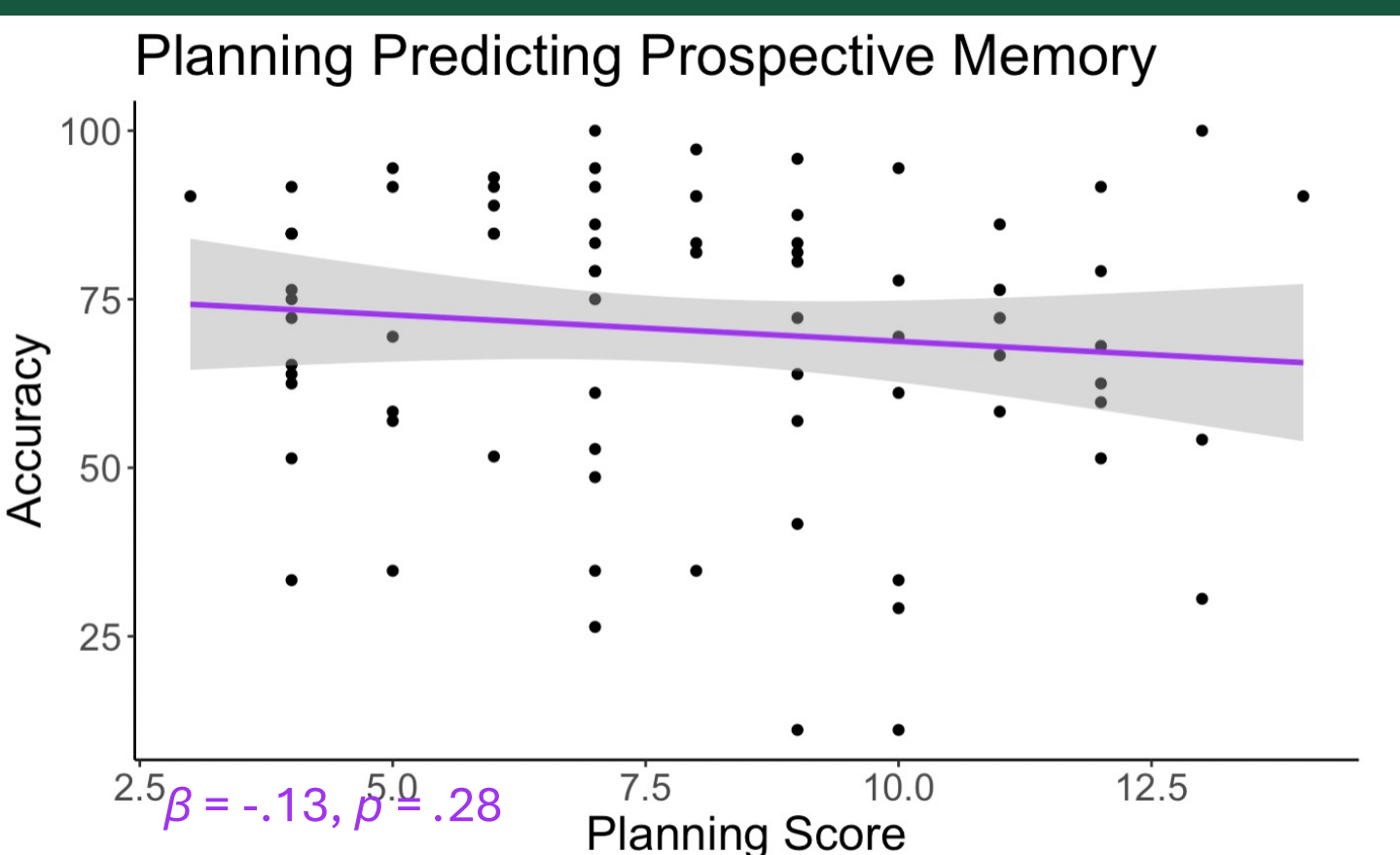
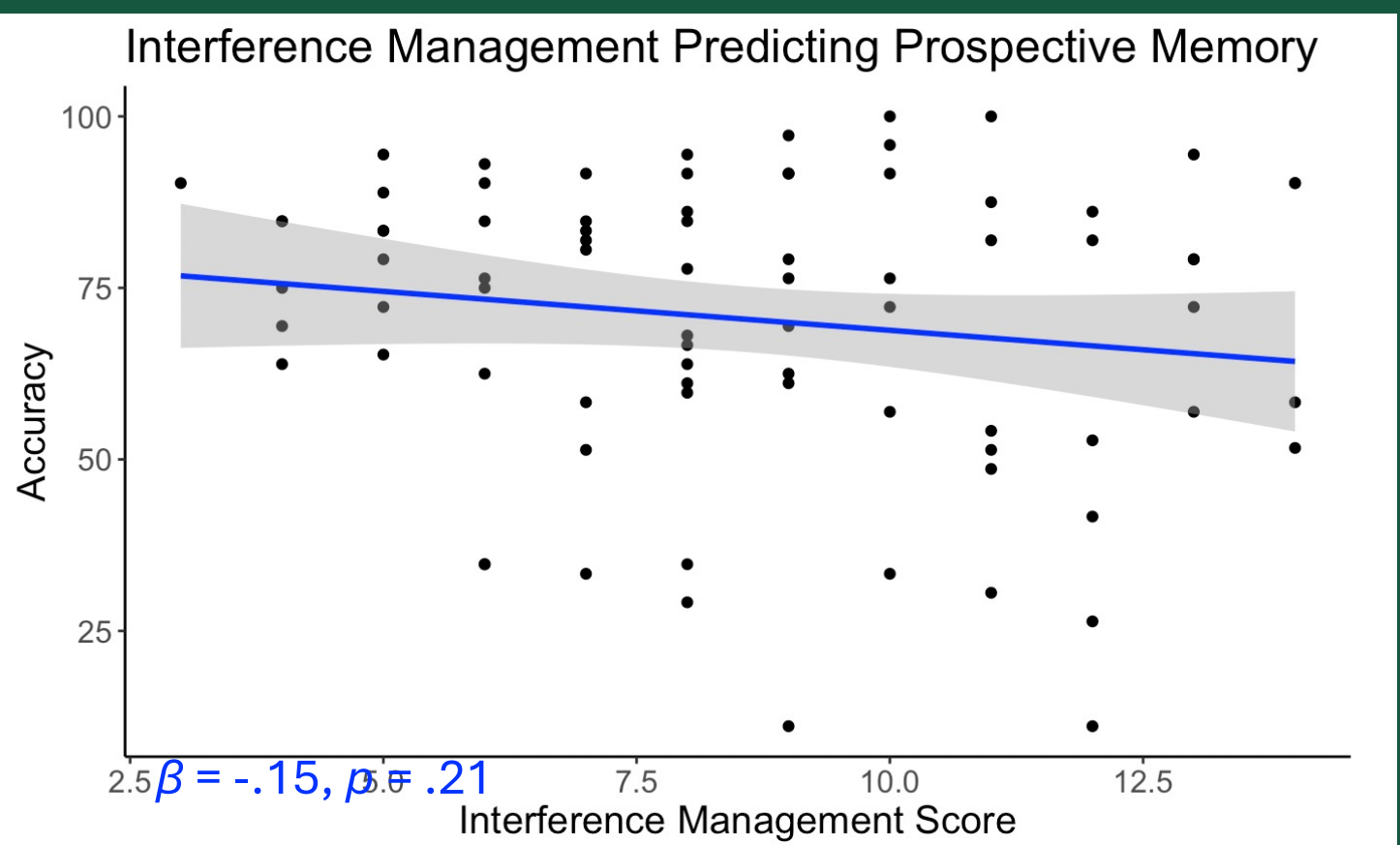
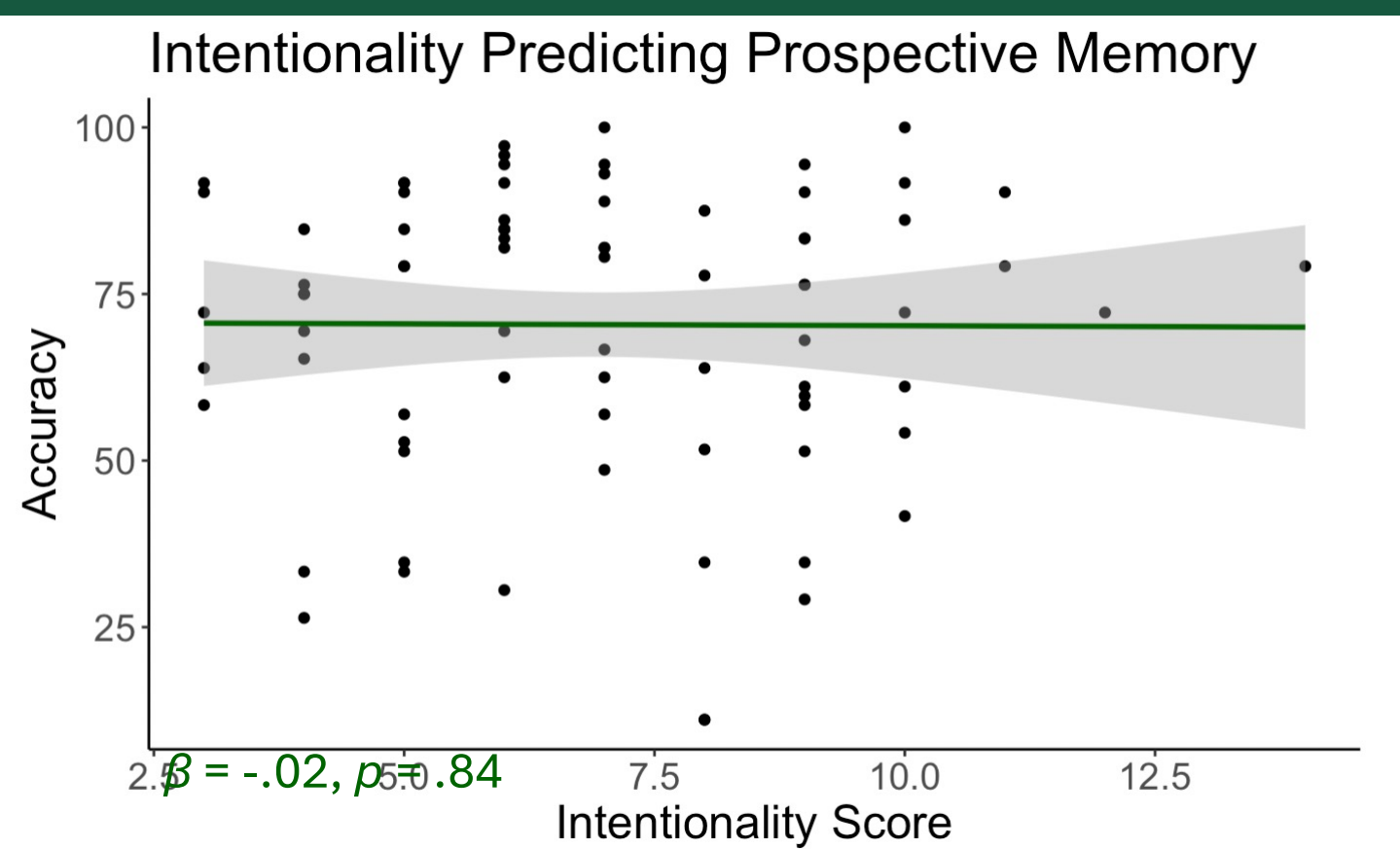
- Higher inhibition scores were significantly associated with lower PM accuracy at a trend level ($\beta = -.20, p = .094$).

- All analyses controlled for age and gender.



Results

- No other factors were significantly related to PM accuracy.



Conclusions

- Inhibition impairment may negatively predict PM performance long before the intended action needs to be performed.
 - Inhibition may block monitoring functions of EF, preventing intentions from being remembered.
- Our results, following the PAM model, suggest that the PM processes may need to be uninterrupted to be committed to memory.
- This study provides support for the idea that only some components of EF are involved in PM.
- Students can better understand the limitations of their own memory and develop more effective compensation strategies.

Future Directions

- Further parse cognitive components of EF to examine other factors potentially relating to PM.
- Examine if the relationship between EF cognitive components and PM is altered in various psychopathologies (e.g., anxiety, depression).
- Incorporate electroencephalogram to see if neural activity for PM varies with the EF factors.
- Examine a more comprehensive measure of Inhibition given that the current measure included only three items

Acknowledgements

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